Network Threats and Mitigation & Physical Security and Risk

1.Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

5 Common Network Threats with Real-World Examples

No.	Network Threat	Real-World Example	Prevention
1	Phishing	An Instagram user receives a fake account verification link that steals their login credentials.	Verify links and enable two-factor authentication (2FA).
2	DDoS Attack	An attacker floods a payment service such as Paytm with excessive requests, making it temporarily unavailable.	Use DDoS protection, traffic filtering, and rate limiting.
3	Man-in-the-Middle (MITM)	A user connects to a malicious public Wi-Fi hotspot, allowing an attacker to attempt to intercept their communication with a website.	Use HTTPS, avoid untrusted Wi-Fi, and use a trusted VPN when appropriate.
4	Malware	A WhatsApp user downloads a malicious APK shared through a message, potentially exposing personal data.	Download apps from trusted sources and keep security software updated.
5	Password Attack	An attacker uses leaked passwords from another website to attempt to access someone's Instagram account.	Use unique, strong passwords and enable 2FA.

2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.

attempts it took to guess the correct PIN.

Python Program: Brute-Force Attack Simulation

This program simulates a brute-force attack by checking every possible 4-digit PIN, from 0000 to 9999, until it finds the preset PIN. It also counts and displays the number of attempts required.

```
# Preset PIN for simulation

correct_pin = "4826"

# Initialize attempt counter

attempts = 0
```

```
# Try all possible 4-digit PINs
for i in range(10000):
    guess = f"{i:04d}"
    attempts += 1
    if guess == correct_pin:
        print("PIN guessed successfully:", guess)
        print("Number of attempts:", attempts)
        break
```

Sample Output

PIN guessed successfully: 4826

Number of attempts: 4827

Explanation

Preset PIN: The correct PIN is stored as "4826".

Generate guesses: The for loop checks every PIN from 0000 to 9999.

Count attempts: The attempts variable increases by one for each guess.

Compare PINs: When the guessed PIN matches the correct PIN, the program displays the PIN and total attempts.

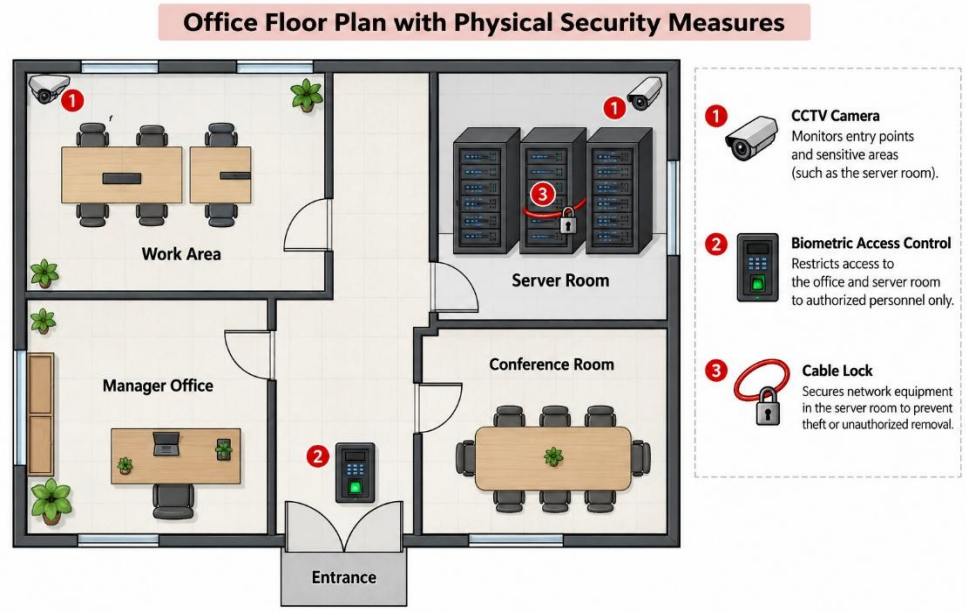
Stop the loop: The break statement stops execution once the correct PIN is found.

3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Physical Security Risks in a Server Room

No.	Physical Security Risk	Mitigation Step
1	Unauthorized Access: An unauthorized person enters the server room and steals or damages equipment.	Install biometric access control, security locks, and CCTV cameras.
2	Fire: Electrical faults or overheating can cause a fire and damage servers.	Install fire detection systems and suitable clean-agent fire suppression systems.
3	Power Failure: A sudden power outage can shut down servers and interrupt services.	Use an Uninterruptible Power Supply (UPS) and backup generators.
4	Overheating: Excessive heat can cause servers to malfunction or fail.	Install reliable cooling systems and temperature monitoring sensors.

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).



5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

Checklist: Network and Physical Security for a College Computer Lab

Objective: To identify seven important actions a network administrator should take to protect a college computer lab from network threats and physical security breaches.

No.	Checklist Action	Specific Implementation for a College Lab
1	Configure Firewalls	Enable firewall protection on lab computers and the college network to block unauthorized connections.
2	Install Antivirus Software	Keep antivirus software updated on all lab computers and schedule regular malware scans.
3	Use Strong Authentication	Require strong passwords for administrator accounts and use multi-factor authentication (MFA) where possible.
4	Restrict Physical Access	Lock the server/network equipment room and allow access only to authorized IT staff.
5	Install CCTV Cameras	Monitor lab entrances, server rooms, and areas containing network switches and routers.
6	Secure Network Equipment	Use locked racks and cable locks to prevent theft or unauthorized removal of routers, switches, and other devices.

7	Update and Back Up Systems	Regularly install security updates, back up important college data, and test recovery procedures.

Review and Editing

The checklist has been adapted to the college computer lab environment by including shared student computers, administrator accounts, restricted server-room access, and protection of networking equipment.

- During lab hours: Ensure students cannot access administrator settings or install unauthorized software.
- After lab hours: Lock the lab and equipment room, and check that CCTV and physical security measures are functioning.
- Regularly: Review firewall logs, update software, scan for malware, and verify backups.